Human Ancestors



Transition to Modern Humans

Homo heidelbergensis / antecessor

- Fossils
  - found in Africa (dated to 1 mya)
  - Found in Europe (dated to 500 000 years ago)
- Features between *H. erectus* and *H. sapiens*
  - Possible common ancestor of Humans and Neanderthals
  - Modern skull contours and teeth
- Not enough fossil evidence to fully understand relationships between different *Homo* species.
  - Family tree has more branches than at first thought – just all extinct.

Neanderthals

- First fossils found in 1856 in Neander Valley, Germany
- Many more fossils found since
- Side branch along pathway to modern humans – we share a common ancestor
- DNA shows a distinct biological species:

  Homo neanderthalensis
- Existed in europe during the last ice age – adapted
- Neanderthals and *Homo sapiens* coexisted in europe, may have inter-bred

Neanderthals



- Features include:
  - Big faces
  - Low but large skulls – back of skull drawn out into “bun”
  - Heavy brow ridges
  - Lack of defined chin
  - Cheeks swept back
  - Prognathic jaw
  - Likely to have very prominent, wide nose – adaptation for cold, dry
  - Solid frame
  - Probably walked, ran and used hands in way very similar to modern humans



Modern Humans

- 1868: fossils found at Cro-Magnon in France
  - Five people
  - Animal bones
  - Sea shell necklaces
  - Stone tools
  - Part of widespread population in Europe from 40 000 years ago to 12 000 years ago.
  - Members of *Homo sapiens*
    - Less prominent brow ridges
    - Shorter, rounder skulls
    - Flatter face
    - Smaller jaw, smaller teeth
    - Chin
    - Large brain
    - Well separated eye sockets

